

Empirical Industrial Organization, Spring 2026

(Profs: Paola Bordón and Jose Manuel Paz y Miño)

Homework 3: Coffee Demand, Conduct, and Counterfactuals

Instructions

- Submit your homework by June 16th at 8pm.
- Your submission must include: (i) Python scripts or notebooks, (ii) a short report with tables, figures, and economic interpretation, and (iii) a README explaining how to run your code.
- Send the homework to schacon@fen.uchile.cl and jpazm@fen.uchile.cl.
- You are encouraged to work in groups of two and collaborate. Each group must submit one report.
- Use Python. You may use `numpy`, `pandas`, `scipy`, `statsmodels`, and `matplotlib`.
- Enjoy!

Overview

This homework studies the international coffee market using the empirical logic of Igami (2015). First, you will estimate inverse demand for coffee. Second, you will use the estimated inverse-demand system to recover supply-side objects under alternative assumptions about competition. Third, you will simulate counterfactual market equilibria under different market structures and demand/cost shocks. The goal is not only to compute equilibria, but also to understand how demand, market power, and leader-follower strategic behavior allocate gains and losses across exporting countries.

1 Data preparation

You are given data on the international coffee market. The dataset contains monthly observations from 1960 to 2006 and includes at least the following objects:

- P_t : international coffee bean price (`price_y`);
- q_{it} : exports of country i (`qe_yc`);
- X_t : importing countries' real GDP or demand shifter (`importGDP_y`);
- Z_t : tea price (`teaPrices_y`);

- W_t : weather-shock instruments (tmax_ye, tmin_ye);
- mc_{it}^{obs} : observed farm-gate or domestic wholesale price, when available (farm_prices_yc);
- f_y : world fertilizer price (fert_y).

Obtain world aggregate exports by summing exported quantities across countries. Call this variable Q_t .

2 Demand estimation

Transform the database so that each row is a year. Use annual variation in the demand estimation.

- (a) Estimate the following linear inverse-demand equation by OLS and by IV:

$$P_t = \alpha_0 + \alpha_Q Q_t + \alpha_X X_t + \alpha_Z Z_t + u_t. \quad (1)$$

In the IV specifications, instrument Q_t using:

- world fertilizer price (fert) and temperature shocks (tmax, tmin) –spec 1–; and
- only temperature shocks (tmax, tmin) –spec 2–.

Interpret the signs and magnitudes of the estimated coefficients.

- (b) Estimate the following log-linear inverse-demand equation by OLS and by IV:

$$\log P_t = \beta_0 + \beta_Q \log Q_t + \beta_X \log X_t + \beta_Z \log Z_t + v_t. \quad (2)$$

In the IV specifications, instrument $\log Q_t$ using the excluded instruments. Use $\log f_t$ if fertilizer prices are strictly positive, and keep temperature shocks in levels or as indicators.

- (c) For each specification, compute and plot the implied price elasticity of demand over time. Discuss whether market demand is elastic or inelastic during the sample period.

Deliverables. Report a table of demand estimates and a figure with elasticities over time. Briefly explain which linear inverse-demand estimate you will use in the supply-side sections.

3 Marginal-cost and markup recovery under alternative conduct assumptions

This section studies how demand estimates map into supply-side primitives. Use all exporting countries in the supply-side calculations whenever the data allow it. For the supply-side recovery and counterfactuals, use spec 2 from the linear inverse-demand specification in equation (1).

- (a) Assume all countries compete as Cournot quantity setters in a homogeneous-good market. Recover the implied marginal cost for each country-year observation.

- (b) Assume the supply side consists of a fixed set of leader and follower countries. The leaders are Brazil, Vietnam, Colombia, and Indonesia. Leaders choose quantities simultaneously in the first stage. Followers observe the aggregate quantity chosen by leaders and then choose quantities simultaneously in the second stage. The market price is determined after all quantities are chosen. Recover the implied marginal cost for each country-year observation under this Stackelberg timing. Appendix A provides the abstract first-order conditions.

Deliverables. Report one figure for each leader country comparing recovered marginal costs under Cournot and Stackelberg. Report one figure for each leader country comparing recovered markups under Cournot and Stackelberg. Explain the difference between the Cournot and Stackelberg approaches.

4 Counterfactual simulations

Assume countries compete according to the Stackelberg timing in Section 3 and use spec 2 from the linear inverse-demand specification. If these modelling assumptions returned a negative marginal cost m_{yc} , replace the marginal cost with the percentile 25 of the non-negative marginal cost recovered in the given year.

For each counterfactual, compute:

- equilibrium price;
- total quantity;
- country-level quantities and market shares;¹
- country-level profits;
- consumer surplus;
- total surplus.

Compare each counterfactual with the baseline Stackelberg equilibrium and, when useful, with the observed data.

Let Ω denote an ownership or coordination matrix. In the baseline case, each country is independently owned, so Ω is the identity matrix. If $\Omega_{ij} = 1$ for $i \neq j$, countries i and j internalize the effect of their quantity choices on each other's profits. If $\Omega_{ij} = 0$, they do not internalize those effects.

Each counterfactual below should be interpreted as an economic question, not only as a numerical exercise.

- (a) **Counterfactual 1: Value of follower cooperation.** Assume that Peru, Honduras, and Mexico form a cooperative bloc among followers. Modify the relevant entries of the follower ownership matrix Ω^F so that these countries internalize the effect of their quantities on each other's profits. Recompute the equilibrium.

¹We are not allowing for an outside-good option, so inside shares sum to 1.

Compute the total gain from cooperation for the bloc,

$$\Delta \Pi_t^{PHM} = \left(\Pi_t^{Peru} + \Pi_t^{Honduras} + \Pi_t^{Mexico} \right)^{coop} - \left(\Pi_t^{Peru} + \Pi_t^{Honduras} + \Pi_t^{Mexico} \right)^{base}.$$

Interpret this gain as the maximum joint amount these countries would be willing to pay to coordinate. Also report each country's individual profit change. Which country is willing to pay more as percentage of its base profit.

- (b) **Counterfactual 2: Bad crop among leaders.** Suppose leader countries suffer a bad crop or adverse productivity shock that raises their marginal costs by 5%. Recompute the equilibrium.

Report how leader quantities, follower quantities, price, leader profits, and follower profits change. Do followers benefit from the leaders' cost increase? Explain the mechanism using the leader-follower structure.

- (c) **Counterfactual 3: Tea becomes more competitive.** Suppose tea becomes a stronger competitor to coffee because the tea price falls by 5%, so that $Z_t^{cf} = 0.95Z_t$. Recompute the equilibrium.

Report who loses more from the demand-side shock: leaders or followers. Compare both absolute and percentage changes in country-level profits. Discuss whether the effect is driven mostly by lower prices, lower quantities, or both.

- (d) **Counterfactual 4: Value of leadership.** Assume that Indonesia stops being a leader and becomes a follower. Recompute the equilibrium.

Compute Indonesia's change in profit,

$$\Delta \Pi_t^{Indonesia} = \Pi_t^{Indonesia, follower} - \Pi_t^{Indonesia, leader}.$$

Interpret this object as a measure of the strategic value of leadership. Also report how the other leaders and followers are affected.

Deliverables. For each counterfactual, report one table comparing the baseline and counterfactual equilibria. Include average price, total quantity, consumer surplus, total producer profits separated between leaders and followers, and total surplus. Add one figure showing how country-level profits change across counterfactuals for all countries. In the text, answer the economic question attached to each counterfactual. Also report from the recovered quantities how many are negative, if any.

A Stackelberg quantity game with heterogeneous marginal costs

This appendix derives the first-order conditions for a two-stage homogeneous-good quantity game with N_L leaders and N_F followers. The derivation is written in abstract form so that you can substitute the estimated inverse-linear demand from equation (1).

Let $\mathcal{L}$ denote the set of leaders and $\mathcal{F}$ the set of followers. Country i has constant marginal cost c_i and chooses quantity $q_i \geq 0$. Aggregate quantity is

$$Q = \sum_{i \in \mathcal{L}} q_i + \sum_{f \in \mathcal{F}} q_f.$$

The inverse demand is $P(Q; X, Z)$, with $P_Q(Q; X, Z) < 0$. Country i 's operating profit is

$$\pi_i(q) = [P(Q; X, Z) - c_i] q_i.$$

The game has two stages. First, leaders choose quantities simultaneously. Second, followers observe the leaders' quantities and choose quantities simultaneously. The market price is then determined by aggregate quantity.

Follower problem

Given leader quantities q_L , followers choose q_F in the second stage. Let Ω^F be the follower ownership or coordination matrix. The first-order condition for follower $f \in \mathcal{F}$ is

$$G_f(q_F, q_L; c, \Omega^F) \equiv P(Q; X, Z) - c_f + P_Q(Q; X, Z) \sum_{k \in \mathcal{F}} \Omega_{fk}^F q_k = 0. \quad (3)$$

When Ω^F is the identity matrix, each follower behaves independently. When some off-diagonal entries of Ω^F are equal to one, those followers internalize each other's output effects.

Let $G(q_F, q_L; c, \Omega^F)$ collect the follower FOCs. The follower reaction derivatives with respect to leader quantities are obtained from the implicit function theorem:

$$\frac{\partial q_F}{\partial q_L} = - \left(\frac{\partial G}{\partial q_F} \right)^{-1} \left(\frac{\partial G}{\partial q_L} \right). \quad (4)$$

This matrix captures how followers adjust their quantities when leaders change output.

Leader problem

Leaders choose quantities in the first stage, anticipating the follower reaction $q_F(q_L)$. Let

$$R_{f\ell} \equiv \frac{\partial q_f}{\partial q_\ell}, \quad \tau_\ell \equiv 1 + \sum_{f \in \mathcal{F}} R_{f\ell}.$$

The term τ_ℓ is the total effect of leader ℓ 's quantity on aggregate output after followers adjust. If leaders act independently, the first-order condition for leader $\ell \in \mathcal{L}$ is

$$P(Q; X, Z) - c_\ell + P_Q(Q; X, Z) q_\ell \tau_\ell = 0. \quad (5)$$

More generally, if leaders have an ownership or coordination matrix Ω^L , the FOC for leader ℓ is

$$\sum_{m \in \mathcal{L}} \Omega_{\ell m}^L [\mathbf{1}\{m = \ell\} (P(Q; X, Z) - c_m) + P_Q(Q; X, Z) q_m \tau_\ell] = 0. \quad (6)$$

In the homework baseline, you can take Ω^L to be the identity matrix.

Special case: inverse-linear demand

For the preferred linear inverse demand,

$$P(Q; X, Z) = a(X, Z) - bQ, \quad b > 0,$$

we have $P_Q = -b$. The follower FOC becomes

$$a(X, Z) - bQ - c_f - b \sum_{k \in \mathcal{F}} \Omega_{fk}^F q_k = 0. \quad (7)$$

For independent leaders, the leader FOC becomes

$$a(X, Z) - bQ - c_\ell - bq_\ell \tau_\ell = 0. \quad (8)$$

References

Igami, Mitsuru. 2015. "Market Power in International Commodity Trade: The Case of Coffee." *Journal of Industrial Economics* 63(2): 225–248.